A Randomized Double-Blind Study of the Effects of Two Low-Dose Combined Oral Contraceptives on Biochemical Aspects

Report from a Seven-Centred Study

WHO SPECIAL PROGRAMME OF RESEARCH, DEVELOPMENT AND RESEARCH TRAINING IN HUMAN REPRODUCTION

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ABSTRACT

A comparative study of the metabolic effects of two combined oral contraceptive preparations was undertaken in seven WHO Collaborating Centres for Research in Human Reproduction. A total of 847 subjects were randomly allocated to one of two pill groups - norethisterone 1mg/ethinyl estradiol 35ug (NET/EE) or levonorgestrel 150ug/ethinyl estradiol 30ug (LNG/EE). An additional 195 women using an IUD served as a comparison group. Blood samples were taken on admission, and at 3 and 12 months thereafter. Both pills induced changes in fasting and 2-hour glucose, triglycerides, total cholesterol, HDL-cholesterol, bilirubin, alkaline phosphatase, albumin, and total protein, but not aspartate aminotransferase. The most dramatic and probably most clinically important changes were an increase in triglycerides and a decrease in HDL-cholesterol. The NET/EE preparation appeared to induce a greater increase in triglycerides, but no significant difference was found between the two pill preparations with respect to HDL-cholesterol changes.

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INTRODUCTION

Since the development of the estrogen-progestogen combination pill for contraception in the early 1960's, the dosage of steroid in commonly-used pill preparations has decreased markedly, with pills containing 30 to 35ug of ethinyl estradiol or 50 ug of mestranol being increasingly used. The two progestogens currently most commonly used in these "low-dose" preparations are norethisterone and levonorgestrel. Although the side effects of combined oral contraceptives have been studied for the past two decades, most of the information accumulated pertains to the higher dose preparations, and additional information is still needed on the low-dose preparations. This is particularly the case in developing countries, where the use of oral contraceptives continues to increase, but where relatively little research on oral contraceptives has been carried out. Studies of the side effects of low-dose oral contraceptives have been conducted mainly in Europe or North America, where disease patterns and risk factors for disease differ substantially from those in developing countries.

One demonstrated side effect of oral contraceptive use has been changes in metabolic parameters, and concern has increased since an association between serum lipid levels and the risk of cardiovascular disease has been established. However, the precise changes associated with specific preparations have not yet been clearly delineated. In response to requests from Member States for more information on the metabolic side effects of currently-used oral contraceptive preparations, and as part of its general strategy to assess the safety of fertility regulating methods, WHO's Special Programme of Research, Development and Research Training in Human Reproduction initiated a multicentred study to compare the metabolic changes associated with the use of two low-dose oral contraceptive preparations among different population groups. The preparations studied were ethinyl estradiol 35 ug with norethisterone 1000 ug (referred to as "NET/EE pill" in this report) and ethinyl estradiol 30 ug with levonorgestrel 150 ug (referred to as "LNG/EE pill").

METHODS

The study was conducted in seven WHO Collaborating Centres for Research in Human Reproduction, in Szeged, Hungary; Ljubljana, Yugoslavia; Alexandria, Egypt; Chandigarh and Bombay, India; Singapore; and Havana, Cuba. Subjects were recruited from the family planning clinics of major teaching hospitals. Women choosing to use oral contraceptives, who fulfilled the eligibility criteria for the study, were randomly allocated in a double-blind manner to one of two pill groups. Women who chose to use an IUD, and who fulfilled the same eligibility criteria, were included as a comparison group. Women were eligible for the study if they were aged 18 to 39, parous, apparently healthy, not breastfeeding, had not used hormonal contraception within the past 3 months and had never used long-acting injectable or implant contraception, and had no contraindications to the use of either oral contraceptives, or an IUD.

An initial blood sample was obtained from all subjects between days 15 and 26 of the pretreatment cycle, and subsequent blood samples were taken at 3 months and 12 months, again between days 15 and 26 of the menstrual cycle. Each subject also received a physical examination at each of these three visits.

Laboratory procedures

In order to allow comparison of results between the different centres' participants in the study, laboratory procedures were standardized and internal and external quality control systems established (1).

Except for Ljubljana, all of the centres in the study utilized Technicon MT II and Mutilyzer Auto Analyzer systems (Ljubljana used a Technicon Auto Analyzer SMA 6). This equipment, which was designed for the small turnover hospital routine laboratory and for research laboratories, was chosen in the mid-1970's as being the most readily available equipment which could be maintained throughout the world and for which a regular standardized supply of reagents could be obtained. The Mutilyzer was a development of the MT II which became available after the commencement of the study. However, the analytical procedures used were virtually identical.

In order to minimize inter-laboratory variation, the laboratories were supplied, whenever possible, with reagents, standards and calibration material from the same batches. In addition, each laboratory established standard internal quality control methods, the results from which were reviewed at regular intervals, and participated in an external quality control scheme operated for the Special Programme by the WHO Collaborating Centre for Research and Reference Service in Clinical Chemistry in Birmingham, which utilizes the procedures established for the monitoring of most clinical chemical laboratories in the United Kingdom. Regular meetings were held between the laboratory staff of the participating centres to establish and review the methodological and quality control procedures and to discuss any problems encountered in adequately maintaining these procedures and operating the equipment.

Laboratory measurements performed on the blood samples included serum levels of both fasting and 2-hr glucose, triglycerides, cholesterol, HDL-cholesterol (except for Ljubljana), aspartate aminotransferase, bilirubin, alkaline phosphatase, albumin and total protein.

The glucose tolerance test was performed after an overnight fast. Glucose (1.0g per kg body weight) was given orally as a 50% solution, followed by up to 200ml of water. Venous blood specimens were taken at 0 and 120 minutes.

Mean changes in the laboratory values were calculated by averaging individual subject changes from admission to 3 months, and from admission to 12 months. In addition, the proportion of subjects with "abnormal" values was tabulated on admission and at 3 and 12 months. Reference ("normal") values were obtained from the reference laboratory which was responsible for the external quality control scheme (2).

Statistical methods

Differences between study groups and between centres on admission were assessed by means of the X^2 test or one-way analysis of variance. The paired t-test was used to determine the significance of changes in laboratory values in each study group between admission and the 3- and 12-month follow-up visits. Analysis of covariance, with the admission value as the covariate, was used to examine the significance of any differences among the study groups with respect to these changes. The laboratory values are shown as the mean value $\pm$ the standard deviation of the mean and the mean change $\pm$ the standard error of the change.

RESULTS

Continuation rates

From the seven participating centres, a total of 1085 women were originally recruited to the study, of whom 43 were excluded because they were subsequently found not to fulfill the study inclusion criteria (2 women because of unacceptably low hemoglobin values and 41 because of abnormal glucose values on admission). Of the 1042 women who fulfilled all of the admission criteria, 429 discontinued before the end of one year, leaving 613 women who completed the entire 12-month study. In general, women in the two

pill groups were more likely to discontinue before the end of the study than women in the IUD group. The discontinuation rates for the two pill groups were similar within each of the centres. Within each study group, however, the discontinuation rates varied markedly among the different centres. The life-table discontinuation rates at one year were 43.5 for the NET/EE pill group, 47.7 for the LNG/EE pill group, and 25.4 for the IUD group. Table I shows the number of women in each study group on admission, at three months, and at 12 months, by centre.

Women who completed 12 months follow-up were compared with women who discontinued before the end of the study, with respect to both admission characteristics and laboratory values on admission. No consistent significant differences were found.

Admission characteristics

The mean age of the subjects was 26.3 years. Overall, IUD users were significantly older than pill users, but there was no significant difference in age between the two pill groups. Similarly, no significant difference in weight was found between the two pill groups, but IUD users weighed significantly more than pill users. Weight varied significantly among the different centres. Women in Alexandria weighed the most (mean weight 71.1 kg) whereas women in Bombay weighed the least (mean weight 43.5 kg). A similar pattern was seen with weight/height.

Information obtained on drinking habits showed that only in Ljubljana were there a substantial number of subjects who reported drinking alcohol (47%), with a higher proportion in the LNG/EE pill group than in either of the other two groups (63% versus 39% in the NET/EE pill group and 31% in the IUD group). Smoking was reported by a substantial proportion of women in Szeged (20%), Ljubljana (48%), and Havana (37%), but only in Ljubljana were there significant differences in smoking habits between the study groups (IUD users smoked less than pill users).

Laboratory values

Admission values

The results of laboratory measurements and the statistical significance of both the mean change from admission and the differences between study groups are shown in Tables II, III, IV, V and VI. All admission biochemical parameters varied significantly between centres. External quality control data indicated that only for glucose (both fasting and 2-hr) was laboratory bias likely to account for some of the differences observed. From the external quality control results, the triglyceride measurements appeared the least reliable; nevertheless, laboratory variations did not appear to account for the differences between centres. For example, Bombay had consistently lower values as compared to other centres, yet the external quality control indicated a positive bias overall.

In contrast, the differences between the three study groups on admission were minimal. With all centres combined, only aspartate aminotransferase and alkaline phosphatase differed on admission between the three groups, and only alkaline phosphatase differed between the two pill groups, suggesting that randomization was successful.

Glucose

No consistent change was observed in fasting glucose blood levels and the various centres differed significantly in the changes reported, in all three study groups. Some centres showed an increase in fasting glucose, others a decrease. In general, the direction of the change at 3 months was the same at 12 months, within each centre and each pill group. In both pill groups, in

Table I: Number of subjects on admission, at 3 months, and at 12 months, by centre, by study group

Centre	Admission			3 months			12 months		
	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
Szeged	69	65	32	61	58	30	39	34	25
Ljubljana	85	83	36	70	68	33	34	40	29
Alexandria	89	87	48	88	82	44	66	68	33
Chandigarh	48	50	25	27	38	21	6	20	14
Bombay	28	20	12	22	14	7	17	12	10
Singapore	45	50	31	36	46	29	29	29	23
Havana	65	63	11	55	54	5	37	39	1
Total	429	418	195	359	360	169	228	242	135

Table II: Fasting and 2-hour glucose values on admission, and change at 3 months and 12 months (adjusted for admission value) by study group, by centre

	Glucose (mg/100ml) (fasting)			Glucose (mg/100ml((2-hr)		
	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
Szeged						
Adm	96.4 ± 9.3	98.2 ± 8.7	98.3 ± 8.0	100.5 ± 16.6	101.6 ± 17.2	99.0 ± 14.6
0-3 m	-0.0 ± 0.9	+0.5 ± 0.9	-0.8 ± 1.3	+8.0 ± 2.7**	+5.3 ± 2.8	+0.4 ± 3.8
0-12 m	-0.1 ± 1.2	-1.7 ± 1.3	-0.1 ± 1.4	+3.8 ± 2.7	+3.0 ± 2.9	-5.2 ± 3.3
Ljubljana						
Adm	85.5 ± 8.2	85.1 ± 7.7	85.3 ± 6.6	83.8 ± 14.4	82.5 ± 13.0	85.5 ± 15.2
0-3 m	-2.5 ± 0.9**	-2.0 ± 0.9*	-0.1 ± 1.3	+0.1 ± 2.7	+0.6 ± 2.6	+3.7 ± 4.0
0-12 m	-1.2 ± 1.3	-1.2 ± 1.2	-2.0 ± 1.4	+0.3 ± 3.3	+7.7 ± 3.1 *	-4.0 ± 3.7
Alexandria						
Adm	92.6 ± 11.1	92.6 ± 10.0	92.7 ± 8.6	104.2 ± 14.2	103.4 ± 13.7	105.5 ± 13.8
0-3 m	+0.4 ± 1.1	+0.8 ± 1.1	-0.5 ± 1.5	+7.0 ± 1.4***	+7.2 ± 1.5***	+0.4 ± 2.0
0-12 m	+3.8 ± 0.9***	+3.4 ± 0.8***	-0.8 ± 1.2	+10.1 ± 1.3***	+8.3 ± 1.3***	-0.5 ± 1.9
Chandigarh						
Adm	91.9 ± 11.3	94.2 ± 10.0	94.6 ± 9.6	99.8 ± 16.2	100.5 ± 16.4	101.2 ± 19.0
0-3 m	+0.7 ± 2.3	+0.5 ± 2.0	-0.0 ± 2.6	+15.9 ± 4.2***	+16.2 ± 3.5***	+6.4 ± 4.7
0-12 m	-2.1 ± 8.3	-8.1 ± 4.7	+4.2 ± 5.4	+16.6 ± 10.8	+11.2 ± 5.9	-4.5 ± 7.3
Bombay						
Adm	93.3 ± 12.2	95.7 ± 12.2	96.2 ± 13.8	95.3 ± 18.0	95.3 ± 16.0	99.6 ± 15.9
0-3 m	-2.7 ± 2.4	-5.3 ± 2.9	+4.4 ± 4.1	+1.8 ± 5.2	+12.1 ± 6.2	+8.3 ± 8.9
0-12 m	-3.6 ± 1.9	-1.3 ± 2.3	+9.4 ± 2.5***	+10.5 ± 5.4	+27.1 ± 6.1***	+23.4 ± 7.0**
Singapore						
Adm	87.3 ± 8.6	84.2 ± 9.0	82.4 ± 10.4	92.3 ± 15.7	93.9 ± 19.0	93.7 ± 17.9
0-3 m	-7.0 ± 1.6***	-2.9 ± 1.4*	-0.0 ± 1.8	+15.2 ± 3.8***	+8.9 ± 3.4*	+4.7 ± 4.4
0-12 m	-5.4 ± 1.7**	-3.4 ± 1.7	+0.8 ± 1.9	+19.0 ± 3.8***	+13.1 ± 3.9***	+7.7 ± 4.5
Havana						
Adm	76.7 ± 9.0	76.0 ± 8.6	78.8 ± 11.2	79.6 ± 13.3	79.8 ± 13.1	80.9 ± 10.3
0-3 m	-2.3 ± 1.5	-2.0 ± 1.5	+1.9 ± 4.9	+4.0 ± 2.2	+6.1 ± 2.2**	-14.1 ± 7.1
0-12 m	+0.2 ± 1.5	+1.7 ± 1.5	-	+9.1 ± 4.0*	+9.3 ± 3.9*	-
All centres combined						
Adm	88.9 ±11.6	88.8 ±11.6	90.3 ±10.9	94.1 ± 17.6	94.2 ± 17.8	97.2 ± 17.1
0-3 m	-1.6 ± 0.5**	-1.1 ± 0.5*	+0.5 ± 0.8	+6.9 ± 1.1***	+6.8 ± 1.1***	+ 2.9 ± 1.6
0-12 m	-0.3 ± 0.7	-0.3 ± 0.6	+1.1 ± 0.9	+8.9 ± 1.3***	+9.2 ± 1.3***	+1.5 ± 1.7

* $p < 0.05$; ** $p < .01$; *** $p < .001$

Table III: Triglycerides, total cholesterol, and HDL-cholesterol values on admission, and change at 3 and 12 months (adjusted for admission values) by study group, by centre

	Triglycerides (mg/100ml)			Total cholesterol (mg/100ml)			HDL-cholesterol (mg/100ml)		
	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
Szeged									
Adm	67±21	66±27	73±27	186±36	193±36	184±36	49±10	51±14	46±11
0-3 m	+28±4***	+18±4***	+8±3*	+8±3*	+4±3	-7±5	-4±1**	-4±1***	-3±2
0-12 m	+38±6***	+27±6***	+0±8	+18±4***	+14±4**	-1±5	-5±1***	-4±2*	-1±2
Ljubljana									
Adm	82±24	86±25	85±23	182±34	184±34	184±27			
0-3 m	+20±5***	+12±5*	-6±7	+3±3	+3±3	+2±5			
0-12 m	+22±6***	+1±5	-16±6	+2±4	-5±4	-10±4			
Alexandria									
Adm	100±33	100±30	96±33	188±37	187±42	194±37	42±12	43±14	46±11
0-3 m	+13±4***	+7±4	+2±6	+2±3	-3±3	2±4	-1±1	-3±1***	+1±1
0-12 m	+22±5***	+12±5*	-5±7	+10±4**	-6±4	-2±5	-4±1	-1±1	+0±2
Chandigarh									
Adm	93±30	93±30	82±28	169±30	173±31	182±34	47±12	45±13	45±12
0-3 m	+38±8***	+5±7	+22±10*	+11±5*	+3±4	-8±6	-2±2	-5±2*	+4±2
0-12 m	+43±11***	+11±6	+14±8	+22±13	+31±8***	+5±4	-9±11	+14±8	-16±11
Bombay									
Adm	72±23	67±22	77±39	160±28	161±27	160±30	43±11	44±15	44±9
0-3 m	+25±6***	+15±7*	-1±10	+13±5**	+1±6	-4±9	-6±2*	-8±3*	-3±4
0-12 m	+34±7***	+6±8	+4±9	+6±6	-0±7	+4±7	-5±2*	-2±3	-2±3
Singapore									
Adm	89±22	91±29	85±34	192±25	202±32	196±33	42±9	42±11	42±10
0-3 m	+31±6***	+6±5	+1±7	-6±4	-12±3***	-6±4	+1±1	-2±1	+0±1
0-12 m	+26±7***	+8±7	+4±8	-1±4	-9±4*	-5±4	+5±2	+1±2	+1±2
Havana									
Adm	78±25	72±22	71±20	183±36	171±31	176±21	48±13	52±17	48±9
0-3 m	+21±4***	+6±4	-17±14	+6±4	-3±4	+2±12	-1±2	-4±2	-8±5
0-12 m	+35±5***	+4±5		10±5	+7±5		-6±2***	-11±2***	
All centres combined									
Adm	83±28	83±29	84±30	182±34	184±36	186±34	46±11	47±14	45±11
0-3 m	+23±2***	+10±2***	+2±3	+4±1**	-1±1	-3±2	-2±1**	-4±1***	-0±1
0-12 m	+29±2***	+10±2***	-2±3	+9±2***	+2±2	-2±2	-2±1*	-3±1**	+0±1

* $p < 0.05$; ** $p < .01$; *** $p < .001$

Table IV: Aspartate aminotransferase, bilirubin, and alkaline phosphatase values on admission, and change at 3 and 12 months (adjusted for admission values) by study group, by centre

	Aspartate aminotransferase (mU/ml)			Bilirubin (mg/100ml)			Alkaline phosphatase (mU/ml)		
	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
Szeged									
Adm	25.9 ± 7.5	24.8 ± 4.8	25.8 ± 7.2	0.7 ± 0.22	0.63 ± 0.21	0.66 ± 0.34	63.5 ± 15.4	59.1 ± 18.8	64.0 ± 15.1
0-3 m	+1.0 ± 1.3	+0.4 ± 1.3	+2.8 ± 1.8	-0.1 ± 0.03**	-0.05 ± 0.03	-0.01 ± 0.04	-10.5 ± 1.1***	-12.9 ± 1.1***	-0.0 ± 1.5
0-12 m	-1.2 ± 1.4	+1.1 ± 1.5	-1.5 ± 1.7	-0.1 ± 0.03***	-0.06 ± 0.03*	-0.01 ± 0.04	-10.2 ± 1.5***	-10.7 ± 1.6***	-1.5 ± 1.8
Ljubljana									
Adm	15.3 ± 5.6	14.9 ± 4.2	17.0 ± 5.7	0.55 ± 0.28	0.52 ± 0.25	0.55 ± 0.25	55.9 ± 19.0	52.8 ± 14.9	62.6 ± 14.9
0-3 m	-1.6 ± 0.8*	-2.3 ± 0.8**	-4.0 ± 1.1***	-0.04 ± 0.03	-0.02 ± 0.03	-0.01 ± 0.04	-11.3 ± 1.1***	-11.1 ± 1.1***	-5.1 ± 1.6**
0-12 m	-5.4 ± 0.6***	-5.9 ± 0.5***	-6.5 ± 0.6***	-0.10 ± 0.03**	-0.06 ± 0.03	-0.04 ± 0.04	-13.3 ± 1.3***	-8.8 ± 1.3***	-4.4 ± 1.5**
Alexandria									
Adm	25.9 ± 8.6	26.9 ± 8.5	27.3 ± 8.3	0.41 ± 0.18	0.45 ± 0.20	0.43 ± 0.26	71.6 ± 26.3	67.1 ± 24.6	68.6 ± 16.1
0-3 m	+0.9 ± 1.2	+2.0 ± 1.2	+0.2 ± 1.6	+0.00 ± 0.02	+0.04 ± 0.02	-0.04 ± 0.03	-4.1 ± 2.0*	-4.2 ± 2.1*	+0.4 ± 2.9
0-12 m	+1.5 ± 1.1	+4.4 ± 1.1***	+0.8 ± 1.5	-0.02 ± 0.02	-0.01 ± 0.02	-0.06 ± 0.03*	-3.3 ± 2.1	-2.6 ± 2.1	+0.3 ± 3.0
Chandigarh									
Adm	34.7 ± 7.8	33.7 ± 9.1	33.9 ± 8.4	0.44 ± 0.20	0.47 ± 0.23	0.39 ± 0.22	91.0 ± 24.6	84.7 ± 18.0	93.5 ± 27.4
0-3 m	-4.9 ± 2.0*	-1.4 ± 1.6	+1.3 ± 2.1	+0.05 ± 0.04	-0.06 ± 0.03	-0.05 ± 0.05	-10.6 ± 3.3**	-17.6 ± 2.8***	-4.2 ± 3.8
0-12 m	-8.6 ± 5.5	-5.2 ± 2.5*	-0.1 ± 3.0	-0.07 ± 0.06	-0.11 ± 0.03**	-0.04 ± 0.04	-7.4 ± 6.0	-12.2 ± 3.3***	-0.6 ± 3.7
Bombay									
Adm	25.6 ± 3.7	25.6 ± 7.0	29.6 ± 6.2	0.61 ± 0.23	0.55 ± 0.17	0.48 ± 0.17	84.5 ± 19.3	79.5 ± 30.0	82.3 ± 25.8
0-3 m	-2.1 ± 1.2	-1.4 ± 1.6	+0.1 ± 2.2	-0.10 ± 0.04*	-0.10 ± 0.05	-0.17 ± 0.07*	-13.5 ± 2.8***	-11.5 ± 3.5**	-11.3 ± 4.8*
0-12 m	+2.5 ± 2.4	+6.4 ± 2.8*	+4.4 ± 3.1	-0.13 ± 0.04**	-0.04 ± 0.05	-0.21 ± 0.06***	-16.2 ± 4.6***	-11.9 ± 5.4*	-14.7 ± 6.0*
Singapore									
Adm	22.5 ± 6.0	22.1 ± 8.1	26.1 ± 8.7	0.81 ± 0.34	0.77 ± 0.29	0.84 ± 0.27	68.8 ± 20.7	63.4 ± 16.8	66.8 ± 20.4
0-3 m	-2.7 ± 1.0**	-1.1 ± 0.8	-2.4 ± 1.1*	-0.14 ± 0.03***	-0.12 ± 0.03***	+0.01 ± 0.04	-15.7 ± 1.7***	-12.1 ± 1.5***	-1.8 ± 1.9
0-12 m	-1.2 ± 1.6	-2.8 ± 1.6	-4.5 ± 1.5*	-0.09 ± 0.05	-0.03 ± 0.05	-0.01 ± 0.06	-19.0 ± 1.9***	-17.7 ± 1.9***	-8.6 ± 2.1***
Havana									
Adm	20.7 ± 7.3	19.5 ± 7.7	20.3 ± 5.7	0.50 ± 0.25	0.47 ± 0.24	0.48 ± 0.17	55.6 ± 17.7	48.7 ± 13.6	75.1 ± 27.2
0-3 m	-0.8 ± 1.1	-0.2 ± 1.1	+0.2 ± 3.7	-0.01 ± 0.03	+0.02 ± 0.03	+0.07 ± 0.09	-5.1 ± 1.7**	-2.4 ± 1.7	+3.2 ± 5.6
0-12 m	-0.6 ± 0.9	-0.0 ± 0.9	=	+0.10 ± 0.11	+0.12 ± 0.11	=	-0.4 ± 2.6	+7.5 ± 2.5**	=
All centres combined									
Adm	23.5 ± 8.9	23.1 ± 9.1	25.5 ± 9.0	0.55 ± 0.27	0.54 ± 0.25	0.56 ± 0.30	67.5 ± 23.8	62.5 ± 22.1	70.9 ± 21.7
0-3 m	-1.0 ± 0.5	-0.2 ± 0.5	-0.5 ± 0.7	-0.04 ± 0.01***	-0.03 ± 0.01*	-0.01 ± 0.02	-9.3 ± 0.8***	-9.2 ± 0.8***	-1.7 ± 1.1
0-12 m	-0.9 ± 0.6	+0.1 ± 0.6	-2.1 ± 0.8**	-0.05 ± 0.02*	-0.02 ± 0.02	-0.03 ± 0.03	-8.9 ± 1.0***	-6.2 ± 1.0***	-2.9 ± 1.3*

* $p < 0.05$; ** $p < .01$; *** $p < .001$

Table V: Albumin and total protein values on admission, and change at 3 and 12 months (adjusted for admission values) by study group, by centre

	Albumin (g/100ml)			Total protein (g/100ml)		
	NET/EE	LNG/EE	IUD	NET/EE	LNG/EE	IUD
Szeged						
Adm	4.71 ± 0.29	4.71 ± 0.29	4.63 ± 0.34	7.59 ± 0.41	7.51 ± 0.35	7.54 ± 0.44
0-3 m	-0.21 ± 0.03***	-0.11 ± 0.03***	-0.11 ± 0.05*	-0.12 ± 0.04***	-0.09 ± 0.04*	-0.10 ± 0.05
0-12 m	-0.20 ± 0.04***	-0.15 ± 0.04***	-0.11 ± 0.05*	-0.17 ± 0.05***	-0.09 ± 0.06	-0.05 ± 0.06
Ljubljana						
Adm	4.36 ± 0.31	4.35 ± 0.29	4.41 ± 0.30	7.09 ± 0.37	7.05 ± 0.35	7.20 ± 0.37
0-3 m	-0.13 ± 0.04***	-0.21 ± 0.04***	+0.02 ± 0.05	-0.02 ± 0.05	+0.00 ± 0.05	+0.04 ± 0.08
0-12 m	-0.19 ± 0.05***	-0.11 ± 0.05*	+0.15 ± 0.06**	+0.01 ± 0.06	+0.04 ± 0.06	+0.13 ± 0.07
Alexandria						
Adm	4.25 ± 0.48	4.27 ± 0.48	4.29 ± 0.37	7.34 ± 0.55	7.36 ± 0.44	7.42 ± 0.51
0-3 m	-0.24 ± 0.03***	-0.18 ± 0.04***	+0.01 ± 0.05	-0.11 ± 0.04*	-0.09 ± 0.05*	+0.02 ± 0.06
0-12 m	-0.22 ± 0.44***	-0.16 ± 0.04***	+0.05 ± 0.05	-0.08 ± 0.04*	-0.05 ± 0.04	+0.12 ± 0.05*
Chandigarh						
Adm	4.36 ± 0.36	4.39 ± 0.31	4.44 ± 0.28	7.68 ± 0.41	7.60 ± 0.53	7.88 ± 0.51
0-3 m	-0.17 ± 0.06**	-0.14 ± 0.05**	+0.02 ± 0.07	-0.02 ± 0.07	-0.08 ± 0.06	-0.20 ± 0.08*
0-12 m	-0.17 ± 0.13	-0.11 ± 0.07	+0.05 ± 0.08	-0.23 ± 0.14	-0.09 ± 0.08	+0.11 ± 0.10
Bombay						
Adm	4.33 ± 0.37	4.23 ± 0.36	4.25 ± 0.31	7.58 ± 0.60	7.53 ± 0.37	7.53 ± 0.44
0-3 m	-0.03 ± 0.06	-0.07 ± 0.08	-0.08 ± 0.11	-0.01 ± 0.09	-0.04 ± 0.11	-0.08 ± 0.16
0-12 m	-0.06 ± 0.10	-0.10 ± 0.11	-0.18 ± 0.12	-0.18 ± 0.13	+0.08 ± 0.16	-0.23 ± 0.17
Singapore						
Adm	4.56 ± 0.19	4.55 ± 0.19	4.56 ± 0.23	7.68 ± 0.36	7.72 ± 0.39	7.64 ± 0.34
0-3 m	-0.18 ± 0.03***	-0.07 ± 0.03*	-0.07 ± 0.04	+0.04 ± 0.05	-0.01 ± 0.05	-0.10 ± 0.06
0-12 m	-0.10 ± 0.04*	-0.03 ± 0.04	-0.10 ± 0.05*	+0.01 ± 0.06	-0.10 ± 0.06	-0.16 ± 0.07*
Havana						
Adm	4.26 ± 0.35	4.26 ± 0.34	4.38 ± 0.53	7.23 ± 0.44	7.28 ± 0.51	7.15 ± 0.40
0-3 m	-0.11 ± 0.05*	-0.10 ± 0.05*	+0.17 ± 0.17	-0.18 ± 0.06**	-0.07 ± 0.06	-0.10 ± 0.19
0-12 m	+0.09 ± 0.05	-0.04 ± 0.05	-	-0.04 ± 0.07	-0.01 ± 0.07	-
All centres combined						
Adm	4.40 ± 0.39	4.40 ± 0.38	4.43 ± 0.35	7.40 ± 0.50	7.39 ± 0.42	7.49 ± 0.48
0-3 m	-0.18 ± 0.02***	-0.14 ± 0.02***	-0.01 ± 0.02	-0.08 ± 0.02***	-0.06 ± 0.02**	-0.03 ± 0.03
0-12 m	-0.14 ± 0.02***	-0.11 ± 0.02***	+0.02 ± 0.03	-0.08 ± 0.02**	-0.04 ± 0.02	+0.02 ± 0.03

* $p < 0.05$; ** $p < .01$; *** $p < .001$

Table VI: Significant differences in change of biochemical parameters from admission, NET/EE pill versus LNG/EE pill and pills versus IUDs, at 3 and 12 months[a]

	Fasting glucose		2-hr glucose		Triglycerides		Cholesterol		HDL-cholesterol	
Centre	NETvs.LNG	Pills vs. IUD	NETvs.LNG	Pills vs. IUD	NETvs.LNG	Pillsvs.IUD	Netvs.LNG	Pillsvs.IUD	Net.vs.LNG	Pillsvs.IUD
Szeged				/*	*/	*/***		*/**		
Ljubljana					/**	**/***		**/***		
Alexandria				**/***		/**	/***			***/
Chandigarh					**/*			*/		*/
Bombay		/***			/*					
Singapore		*/*			**/	*/			*/	
Havana				*/	*/***	*/			/*	
All centres combined		*/		*/***	***/***	***/***	*/**	/**		**/

	Aspartate aminotransferase		Bilirubin		Alkaline Phosphatase		Albumin		Total protein	
Centre	NETvs.LNG	Pills vs. IUD	NETvs.LNG	Pills vs. IUD	NETvs.LNG	Pillsvs.IUD	Netvs.LNG	Pillsvs.IUD	Net.vs.LNG	Pillsvs.IUD
Szeged						***/***	*/			
Ljubljana					/*	***/***		**/***		
Alexandria								***/***		/**
Chandigarh						*/		*/		
Bombay										
Singapore				**/		***/***	*/			
Havana					/*					/*
All centres combined						***/**		***/***		

[a] Significance indicated as 3 months/12 months
* $p < 0.05$; ** $p < .01$; *** $p < .001$

all centres, an increase in 2-hr glucose was observed at both 3 months and 12 months. The increase was significant for both pill groups in several centres and with all centres combined. Overall, both pill groups had a significantly greater mean increase in 2-hr glucose, as compared with the IUD group, but the two pill groups did not differ significantly.

All women enrolled in the study had both fasting and 2-hour glucose values within the normal range on admission. At both 3 and 12 months, 8% of the pill users showed impaired glucose tolerance, compared with 4% in the IUD group. By 12 months, one-half of the pill users and one of the IUD users with impaired glucose tolerance at 3 months had discontinued from the study. Approximately one-third of pill users and all but one of the IUD users with impaired glucose tolerance at 3 months showed normal glucose tolerance at 12 months.

Triglycerides

A mean increase in the triglyceride level was observed in all centres among both pill groups. This increase was significantly greater than the change among IUD users in Szeged and Ljubljana at 3 and 12 months, in Singapore and Havana at 3 months, in Alexandria at 12 months and with all centres combined, at both 3 and 12 months. The increase in triglycerides was greater among NET/EE pill users than among LNG/EE pill users in all centres at both visits, the difference being statistically significant when all centres were combined. Reflecting the differences in the change in mean triglyceride values, the proportion of subjects who shifted from normal values on admission, to high values at 3 and 12 months, was greater among NET/EE pill users than among LNG/EE pill users. At 12 months, 23% of NET/EE pill subjects had high triglyceride levels (compared with 6% on admission), whereas only 8% of LNG/EE pill users had high levels at 12 months (compared with 5% on admission). This pattern was observed in each of the centres, at both 3 and 12 months.

Total cholesterol

The mean total cholesterol level increased among NET/EE pill users in all centres except Singapore at 3 months and in all centres at 12 months. In contrast, among LNG/EE pill users, a decrease in the mean cholesterol level was found in 3 centres at 3 months and in 4 centres at 12 months. With all centres combined, no significant change in cholesterol was found among LNG/EE pill users, whereas the increase among NET/EE pill users was both highly significant, and significantly greater than the change among LNG/EE pill users. However, a cholesterol level above the normal range was found only in one subject, at 12 months.

HDL-cholesterol

The mean HDL-cholesterol level decreased among NET/EE pill users at 3 months in 5 of the 6 centres performing the test. At 12 months, 4 centres showed a significant decrease among NET/EE pill users. Among LNG/EE pill users, all centres showed a decrease at 3 months, with 4 of the 6 centres showing a significant decrease. At 12 months, two centres showed a significant decrease and 3 more centres a non-significant decrease among LNG/EE pill users.

With all centres combined, the mean decrease in HDL-cholesterol level was significant at 3 and 12 months in both pill groups, and at 3 months was significantly different from the change in the IUD group. However, there was no significant difference between the two pill groups. The proportion of subjects with an HDL-cholesterol level below the normal range increased (though not significantly) in both pill groups from 15% on admission, to 20% at 12 months.

Aspartate aminotransferase

In general, aspartate aminotransferase decreased among both NET/EE pill and LNG/EE pill users, although the decrease was less consistent among LNG/EE pill users. With all centres combined, no significant change was found in either pill group. There was no significant difference between the change in pill users as compared to IUD users, in any of the centres, or with all centres combined.

Bilirubin

Bilirubin also decreased among pill users in most centres, with 5 centres showing a significant decrease at both 3 and 12 months. No consistent differences were found between NET/EE pill and LNG/EE pill users. With all centres combined, bilirubin decreased significantly at 3 months among both pill groups, and at 12 months among NET/EE pill users. Bilirubin also decreased among IUD users, but not significantly, and there was no significant difference between the change in pill users and IUD users.

Alkaline phosphatase

Except for LNG/EE pill users in Havana at 12 months, all centres reported a decrease in alkaline phosphatase among both NET/EE pill and LNG/EE pill users, at both 3 and 12 months. With all centres combined, a significant decrease was found in both pill groups, which was significantly greater than the decrease in the IUD group at both 3 and 12 months. There was no significant difference in the change, between the two pill groups.

Albumin

Except for NET/EE pill users in Havana at 12 months, all centres reported a mean decrease in serum albumin levels among both NET/EE pill and LNG/EE pill users at both 3 and 12 months, although the decrease was not always significant. With all centres combined, the decrease was highly significant in both pill groups, although small in magnitude, and significantly different from the IUD group, at both 3 and 12 months. No significant difference was found between the two pill groups.

Total protein

Total protein levels also decreased slightly in most centres, for both NET/EE pill and LNG/EE pill groups, although only 3 centres showed significant changes. With all centres combined, the decrease among both pill groups was significant at 3 months for both pill groups and at 12 months for the NET/EE pill groups, although again, small in magnitude. No subjects had a protein level below the normal range.

DISCUSSION

Both oral contraceptive preparations induced changes in the metabolic parameters measured. For many of the parameters, the differences between centres were more marked than the differences between study groups. Although age, weight, weight/height, and all the laboratory values varied significantly between centres on admission, neither age, weight, nor weight/height was consistently correlated with any of the laboratory values at admission or the changes at 3 and 12 months. Similarly, no consistently significant differences in biochemical parameters on admission were seen between smoking and non-smoking women, nor between women who reported drinking alcohol and those who did not. This suggests that other factors, whether genetic/ethnic or environmental, must account for the sometimes striking differences between centres.

No substantial differences between pill groups were found with respect to glucose changes; both pill groups induced a mean increase in 2-hr glucose in all centres, although changes in fasting glucose were not consistent and were of smaller magnitude. Differences between centres were more striking than differences between pill groups, for both fasting and 2-hr glucose values, with respect to both admission values and changes at 3 and 12 months. In general, studies of glucose tolerance with oral contraceptive use have shown few, if any changes, in fasting glucose, and it has been suggested that the small change sometimes seen, particularly in glucose tolerance, is due to a change in insulin response (3,4).

The most consistent change induced by both pills in this study was an increase in triglyceride levels. The increase was consistently greater among NET/EE pill users. The consistency among centres was striking, and the magnitude of the increase would suggest that use of particularly NET/EE pill might induce triglyceride changes of clinical importance. The change in triglycerides did, however, tend to correlate positively with 2-hr glucose value on admission. A number of published studies have also reported an increase in serum triglyceride levels with the use of both pill preparations included in this study, although some have also reported no significant change in triglycerides (3,5,6,7,8,9). However, any association between serum triglycerides and coronary heart disease remains unconfirmed (10).

The pattern of total cholesterol changes in the study was similar to that of triglyceride changes, except that only one subject had an abnormally high cholesterol level during the study. In contrast to both triglycerides and total cholesterol, no substantial difference was seen between the two pill groups in HDL-cholesterol. Both pills induced a decrease in HDL-cholesterol, although not in every centre.

The level of serum cholesterol, and more specifically, HDL-cholesterol, has been shown to be associated with the risk of arterial disease, although the effect of any oral contraceptive-induced changes in HDL-cholesterol on the risk of arterial disease, has yet to be clarified. Nevertheless, a substantial and/or prolonged decrease in HDL-cholesterol would be of potential concern. Consequently, a number of studies have reported on changes in HDL-cholesterol associated with the use of oral contraceptives (3,5,6,7,8,9). Most have reported a decrease in HDL-cholesterol, although the magnitude of the decrease has varied. Both the type of progestogen and the progestogen:estrogen ratio have been suggested as influencing the changes in lipids (11,12), but a consistent difference between low-dose oral contraceptives has not been demonstrated, which is consistent with the findings from this study.

Changes in aspartate aminotransferase and bilirubin differed among centres more than between pill groups. Alkaline phosphatase decreased among both pill groups in all centres, significantly more than among IUD users in many centres. Similarly, albumin decreased among both pill groups in all but one centre. No differences between pill groups were seen for alkaline phosphatase, albumin, or total protein changes.

In summary, both the norethisterone acetate- and the levonorgestrel-containing pills induced changes in all of the metabolic parameters measured, except aspartate aminotransferase. The most dramatic and probably most important changes, in terms of potential adverse effects on the risk of cardiovascular disease, were an increase in triglycerides and a decrease in HDL-cholesterol, consistent with most other studies. The norethisterone-containing preparation appeared to induce a greater increase in triglycerides, but no significant difference was found between the two pills with respect to HDL-cholesterol changes.

REFEREMCES

1. Whitehead, T.P. Quality Control in Clinical Chemistry. Wiley, New York (1977).

2. Bold, A.M. and Wilding, P. Clinical Chemistry Companion. Blackwell Scientific Publications, Oxford (1978)

3. Roy, S., Mishell, D.R. (Jr.), Gray, G., Dozono-Takano, R., Brenner, P.F., Eide, I., de Quattro, V. and Shaw, S.T. Comparison of metabolic and clinical effect of four oral contraceptive formulations and a contraceptive vaginal ring. Am J Obst Gyn 136: 920-931 (1980)

4. Briggs, M. and Briggs, M. Biochemistry and metabolism, carbohydrates. In: Oral Contraceptives. Eden Medical Research Inc., St. Abans, Vermont 181-183 (1981)

5. Powell, M.G., Hedlin, A.M., Cerskus, I., Kakis, G., Prudham, D. and Rosenrot, P. Effects of oral contraceptives on lipoprotein lipids: a prospective study. Obstet Gynecol 63: 764-770 (1984)

6. Larsson-Cohn, U., Fahraeus, L., Wallentin, L. and Zador, G. Effects of the estrogenicity of levonorgestrel/ethinylestradiol combinations on the lipoprotein status. Acta Obstet Gynecol Scand 105: 37-40 (1982)

7. Briggs, M. and Briggs, M. A randomized study of metabolic effects of four low-estrogen oral contraceptives: I. Results after 6 cycles. Contraception 23: 463-471 (1981)

8. Rossner, S., Frankman, O. and Marsk, L. Effects of ethinylestradiol/D-norgestrel combinations on serum lipoproteins. Acta Obstet Gynecol Scand 59: 255-258 (1980)

9. Larsson-Cohn, U., Wallentin, L. and Zador, G. Effects of three different combinations of ethinyl estradiol and levonorgestrel on plasma lipids and high density lipoproteins. Acta Obstet Gynecol Scand 88: 57-60 (1979)

10. Hulley, S.B., Rosenman, R.H., Bawol, R.D. and Brand, R.J. Epidemiology as a guide to clinical decisions. The association between triglyceride and coronary heart didsease. New Eng J Med 302: 1383-1389 (1980)

11. Kay, C.R. Progestogens and arterial disease. Evidence from the Royal College of General Practitioners' Study. Am J Obstet Gynec 142: 762-765 (1982)

12. Wahl, P., Walden, C., Knopp, R., Hoover, J., Wallace, R., Heiss, G. and Rifkind. Effect of estrogen/progestin potency on lipid lipoprotein cholesterol. N Engl J Med 308: 862-867 (1983)